Pedro Zuidberg Dos Martires, Ph.D.

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https://pedrozudo.github.io

Employment

Apr 2022 –	**Örebro University** (Sweden) Postdoctoral Researcher in the MPI Lab funded by WASP
Dec 2020 – Feb 2022	**KU Leuven** (Belgium) Postdoctoral Researcher in the DTAI Lab
Apr 2016 - Nov 2020	**KU Leuven** (Belgium) Research Assistant in the DTAI Lab

Education

Apr 2016 - Nov 2020	**KU Leuven** (Belgium), Department of Computer Science PhD in Computer Science, Artificial Intelligence *From Atoms to Possible Worlds: Probabilistic Inference in the Discrete- Continuous Domain* *Defended on November 25, 2020* *Supervised by Prof. Luc De Raedt*
Aug 2013 - Aug 2015	**University of Amsterdam** (Netherlands), Faculty of Physics Master of Science in Particle Physics
Oct 2012 - Jan 2013	**University of Lisbon** (Portugal) Exchange Student
Oct 2009 - Oct 2012	**University of Vienna** (Austria), Faculty of Physics Bachelor of Science in Physics

Research Interests

Broadly speaking, my research interests lie with artificial intelligence and machine learning, while having a focus on probabilistic programming. My overarching scientific goal is to develop systems that are capable of reasoning and learning with raw sensor data. In order to achieve this, I believe in the necessity of a holistic approach, which is also reflected in my specific research activities – ranging from probabilistic inference with discrete and continuous variables [5, 6, 9, 10], over formal probabilistic logic programming [25, 4, 27], to cognitive robotics [7, 1, 2]. Recently, I have also started exploring neuro-symbolic techniques in order to combine symbolic and subsymbolic representations of the world [3, 16].

Research Visits

Feb 2023 - Mar 2023	Research stay in the **SML Group** (Trento, Italy) *Working on neuro-symbolic AI with with Prof. Andrea Passerini*
Oct 2022	Attendence of the **ELLIIT Focus Period** (Linköping, Sweden) *Two week workshop at the intersection of AI and control theory*
Jan 2022 - Feb 2022	Research stay in the **SML Group** (Trento, Italy) *Working on neuro-symbolic AI with with Prof. Andrea Passerini*
Aug 2018	Research stay at the **MPI Lab** (Örebro, Sweden) *Working on cognitive robotics with Prof. Amy Loutfi*
Jul 2015	Research stay at **CERN** (Geneva, Switzerland) *Data analysis for the LHCf experiment with Prof. Davide Berge*
Jun 2014 - Aug 2014	Summer Student at **CERN** (Geneva, Switzerland) *Data analysis at the Large Hadron Collider supervised by Dr. Pamela Ferrari*

Grants & Awards

Dec 2022	Recipient of a TAILOR Connectivity Fund grant for a two month research visit in Trento, Italy
Jul 2020	Recipient of a Postdoctoral Mandate at the KU Leuven, includes personal stipend for full year of postdoctoral research
Aug 2008	Distinguished paper award at the Junior Mathematical Congress (Jena, Germany)

Teaching

Instructor

Autonomous Robots and ROS, Örebro University
2023
Introductory course to the Robot Operating System for professionals

Summer School on Roboitcs, Örebro University
2022, 2023
Internatinal four week long summer school for undergraduate students learning the to use the Robot Operating System

Teaching Assistant

WASP Artificial Intelligence and Machine Learning, Örebro University
2023
Advanced course for on AI and machine learning for PhD students in the WASP program

Databases, KU Leuven
2020-2021, 2021-2022
Course on databases in the Computer Science Master's program

Uncertainty in Artificial Intelligence, KU Leuven
2016-2017, 2017-2018, 2018-2019, 2019-2020, 2020-2021, 2021-2022
Course on probabilistic artificial intelligence in the Computer Science Master's program

Information Structures and Implications, KU Leuven
2016-2017, 2017-2018, 2020-2021, 2021-2022
Database course for students in the Digital Humanities Master's program

Introduction to Physics, University of Vienna
2011-2012
Introductory physics course on in the Physics Bachelor's program

Mentorship & Advisees

PhD Students

Matthias Möller, Örebro Universty, 2023 –
Co-Supervisor: Prof. Luc De Raedt

Rishi Hazra, Örebro Universty, 2022–
Co-Supervisor: Prof. Luc De Raedt

Lennert De Smet, KU Leuven, 2021 –
Supervisor: Prof. Luc De Raedt

Gabriele Venturato, KU Leuven, 2020 –
Supervisor: Prof. Luc De Raedt

Victor Verreet, KU Leuven, 2019 – 2022
Supervisor: Prof. Luc De Raedt

Master Students

Michiel Baptist, KU Leuven, 2018 – 2019

Pieter-Jan Coenen, KU Leuven, 2018 – 2019

Pieter-Jan Coenen, KU Leuven, 2018 – 2019

Evert Heylen, KU Leuven, 2018 – 2019

Rei Bardhi, KU Leuven, 2018 – 2019

Olivier, Kamers, KU Leuven, 2017 – 2018

	Niels Wéry, KU Leuven, 2016 – 2017
Bachelor Students	**Tibo Van den Eede**, KU Leuven, 2021 – 2022
	Sam Vervaeck, KU Leuven, 2021 – 2022
Interns	**Shani Vanlerberghe**, KU Leuven, Summer 2019
	Ivan Miošić, KU Leuven, Summer 2019 (Erasmus Internship)

Skills

Languages

Native Speaker	Luxembourgish, German
Fluent	English, French
Good Command	Portuguese, Dutch

Soft Skills

Funding	I have experience reporting on external funding (ReGROUND project) and successfully writing grant applications (e.g. Postdoctoral Mandate at the KU Leuven).
Networking	am an active member of the research community and have, for instance, hosted Prof. Robert Peharz and Dr. Vissarion Fisikopoulos at the KU Leuven.
Organizing	I have been involved in organizing various scientific events and workshops, e.g. the biannual Fluffy workshp of the DTAI lab at the KU Leuven or the Summer school on Robotics at the University of Örebro.

Technical Skills

AI/ML Software	ProbLog, PyTorch, Tensorflow, Robot Operating System
Programming languages	Python, C++, Julia, DLang

Professional Activities & Service

Reviewing

I regularly review papers for the following venues:

Journals	Artificial Intelligence Journal
	International Journal of Approximate Reasoning
Conferences	International Conference on Artificial Intelligence and d Statistics (AISTATS)
	Uncertainty in Artificial Intelligence (UAI)
	International Joint Conference on Artificial Intelligence (IJCAI)
	Conference on Artificial Intelligence (AAAI)
	International Joint Conference on Learning and Reasoning (IJCLR)
	International Conference on Probabilistic Programming (ProbProg)
	European Conference on Artificial Intelligence (ECAI)
Workshops	International Workshop on Statistical Relational AI (StarAI)
	Workshop on Tractable Probabilistic Modeling (TPM)

Volunteering

I have participated in the Summer School of Science in Croatia as a project leader. The summer school is a ten day event during which a small group of high school students works out their own research project and present their research to peers. I have had the opportunity to supervise two such projects (2017, 2018).

Publications

Journal Publications

[1] **P. Zuidberg Dos Martires***, N. Kumar, A. Persson, A. Loutfi, and L. De Raedt. "Symbolic Learning and Reasoning with Noisy Data for Probabilistic Anchoring". In: *Frontiers in Robotics and AI* 7 (2020).

[2] A. Persson*, **P. Zuidberg Dos Martires***, L. De Raedt, and A. Loutfi. "Semantic Relational Object Tracking". In: *IEEE Transactions on Cognitive and Developmental Systems* 12.1 (2019).

Conference Publications

[3] L. De Smet*, **P. Zuidberg Dos Martires**, R. Manhaeve, G. Marra, A. Kimmig, and L. De Raedt. "Neural Probabilistic Logic Programming in Discrete-Continuous Domains". In: *Proceedings of the Uncertainty in Artificial Intelligence Conference (UAI)*. 2023.

[4] V. Verreet*, V. Derkinderen, **P. Zuidberg Dos Martires**, and L. De Raedt. "Inference and Learning with Model Uncertainty in Probabilistic Logic Programs". In: *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*. 2022.

[5] P. Morettin*, **P. Zuidberg Dos Martires***, S. Kolb*, and A. Passerini. "Hybrid probabilistic inference with logical and algebraic constraints: a survey". In: *Proceedings of the 30th International Joint Conference on Artificial Intelligence (IJCAI)*. 2021.

[6] V. Derkinderen*, E. Heylen, **P. Zuidberg Dos Martires**, S. Kolb, and L. De Raedt. "Ordering Variables for Weighted Model Integration". In: *Proceedings of the Uncertainty in Artificial Intelligence Conference (UAI)*. 2020.

[7] A. Persson*, **P. Zuidberg Dos Martires**, L. De Raedt, and A. Loutfi. "ProbAnch: a Modular Probabilistic Anchoring Framework". In: *Proceedings of the International Joint Conference on Artificial Intelligence (IJCAI)*. 2020.

[8] S. Kolb*, P. Morettin, **P. Zuidberg Dos Martires**, F. Sommavilla, A. Passerini, R. Sebastiani, and L. De Raedt. "The pywmi Framework and Toolbox for Probabilistic Inference using Weighted Model Integration". In: *Proceedings of the International Joint Conference on Artificial Intelligence (IJCAI)*. 2019.

[9] S. Kolb*, **P. Zuidberg Dos Martires***, and L. De Raedt. "How to Exploit Structure while Solving Weighted Model Integration Problems". In: *Proceedings of the Uncertainty in Artificial Intelligence Conference (UAI)*. 2019.

[10] **P. Zuidberg Dos Martires***, A. Dries, and L. De Raedt. "Exact and Approximate Weighted Model Integration with Probability Density Functions Using Knowledge Compilation". In: *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*. 2019.

Tutorials

[11] P. Morettin, **P. Zuidberg Dos Martires**, S. Kolb, and A. Passerini. *Hybrid Probabilistic Inference with Algebraic and Logical Constraints*. 2022. URL: https://dtai.cs.kuleuven.be/tutorials/wmitutorial.

Workshop Publications

[12] L. De Smet*, E. Sansone, and **P. Zuidberg Dos Martires**. "Differentiable Sampling of Categorical Distributions Using the CatLog-Derivative Trick". In: *Differentiable Almost Everything: Differentiable Relaxations, Algorithms, Operators, and Simulators*. 2023.

[13] V. Derkinderen*, **P. Zuidberg Dos Martires**, S. Kolb, and P. Morettin. "Top-Down Knowledge Compilation for Counting Modulo Theories". In: *Workshop on Counting and Sampling*. 2023.

[14] L. De Smet*, R. Manhaeve, G. Marra, and **P. Zuidberg Dos Martires**. "Tensorised Probabilistic Inference for Neural Probabilistic Logic Programming". In: *The 5th Workshop on Tractable Probabilistic Modeling*. 2022.

[15] G. Venturato*, V. Derkinderen, **P. Zuidberg Dos Martires**, and L. De Raedt. "Towards Tractable Dynamic Decision Making With Circuits". In: *5th Workshop on Tractable Probabilistic Modeling*. 2022.

[16] **P. Zuidberg Dos Martires***. "Neural Semirings". In: *15th International Workshop on Neural-Symbolic Learning and Reasoning (NeSy)*. 2021.

[17] **P. Zuidberg Dos Martires*** and S. Kolb. "Monte Carlo Anti-Differentiation for Approximate Weighted Model Integration". In: *Ninth International Workshop on Statistical Relational AI (StarAI)*. 2020.

[18] O. A. Can*, **P. Zuidberg Dos Martires***, A. Persson, J. Gaal, A. Loutfi, L. De Raedt, D. Yuret, and A. Saffiotti. "Learning from Implicit Information in Natural Language Instructions for Robotic Manipulations". In: *Combined Workshop on Spatial Language Understanding and Grounded Communication for Robotics (SpLU-RoboNLP)*. 2019.

[19] **P. Zuidberg Dos Martires***, A. Dries, and L. De Raedt. "Knowledge Compilation with Continuous Random Variables and its Application in Hybrid Probabilistic Logic Programming". In: *Eigth International Workshop on Statistical Relational AI (StarAI)*. 2018.

[20] L. Antanas, O. A. Can, J. Davis, L. De Raedt, A. Loutfi, A. Persson, A. Saffiotti, E. Unal, D. Yuret, and **P. Zuidberg Dos Martires**. "Relational Symbol Grounding through Affordance Learning: an Overview of the ReGround Project". In: *International Workshop on Grounding Language Understanding (GLU)*. 2017.

Extended Abstracts

[21] V. Verreet*, V. Derkinderen, **P. Zuidberg Dos Martires**, and L. De Raedt. "Inference and Learning with Model Uncertainty in Probabilistic Logic Programs". In: *Proceedings of the International Conference on Logic Programming (ICLP)*. 2022.

[22] **P. Zuidberg Dos Martires***. "Differentiation and Weighted Model Integration". In: *Workshop on Deep Continuous-Discrete Machine Learning (DecodeML)*. 2019.

[23] **P. Zuidberg Dos Martires***, V. Derkinderen, R. Manhaeve, W. Meert, A. Kimmig, and L. De Raedt. "Transforming Probabilistic Programs into Algebraic Circuits for Inference and Learning". In: *Program Transformations for Machine Learning*. 2019.

[24] **P. Zuidberg Dos Martires*** and S. Dumancic. "Reactive Probabilistic Programming". In: *International Conference on Probabilistic Programming (ProbProg)*. 2018.

Preprints

[25] **P. Zuidberg Dos Martires**, L. De Raedt, and A. Kimmig. "Declarative Probabilistic Logic Programming in Discrete-Continuous Domains". In: *arXiv preprint arXiv:2302.10674* (2023).

[26] I. Miosic* and **P. Zuidberg Dos Martires**. "Measure Theoretic Weighted Model Integration". In: *arXiv preprint arXiv:2103.13901* (2021).

Doctoral Thesis

[27] **P. Zuidberg Dos Martires***. "From Atoms to Possible Worlds: Probabilistic Inference in the Discrete-Continuous Domain". PhD thesis. KU Leuven, 2020.